

Joshua Cowell

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected Graduation: May 2027

Liberty University, Lynchburg, VA

GPA: 3.82

Relevant Coursework: Fluid Dynamics, Heat Transfer, Thermodynamics I and II, Differential Equations, Dynamics, Materials Engineering, University Physics II, Statics, Engineering Economy, Physics I and II, Calculus I, II, III

PROFESSIONAL EXPERIENCE

Thermal Hydraulics Engineering Intern

May 2025 – Dec 2025

Framatome Nuclear

- Developed a predictive wear mathematical model in Python for critical reactor components, proving to be over 75% more accurate compared to the previous model.
- Programmed a Python algorithm using a 6-dimensional Xarray to calculate safe reactor operating conditions, reducing computation time from 800 hours to minutes compared to prior methods.
- Performed system-level flow and vibration analysis using AFT Fathom and custom FIV modeling software.
- Awarded \$2,500 for final summer presentation and hired on part-time during the school year.

Chief Systems Engineer

June 2025 – Present

Liberty Rocketry (9th/156 international teams in 2024)

- Led as member captain in the design, testing, and analysis of a high-powered rocket.
- Oversaw all engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems.
- Facilitated communication between internal teams, external vendors, and university faculty to optimize project efficiency.
- Displayed advanced proficiency in engineering software (e.g., SolidWorks, ANSYS, MATLAB) to verify the accuracy and feasibility of all subsystem models and simulations.

Lead Undergraduate Research Assistant

Sep 2024 – Present

Liberty University

- Led a team of 5 undergraduate students to perform split-Hopkinson pressure bar (SHPB) test, including sample preparation, high-speed imaging, electron microscopy.
- Assisted in the design and manufacturing of custom SHPB systems to fit material testing needs.

Senior Structural Engineer

Sep 2023 – June 2025

Liberty Rocketry

- Utilized ANSYS Fluent & Mechanical, RocketPy, OpenRocket, and Onshape for simulations, FEA, and computational modeling, ensuring compliance with competition standards.
- Delivered reports to guarantee the rocket's compliance with competition standards and flight requirements.

Camp Counselor

May 2024 – Aug 2024

Woodlands Camp, GA

- Provided 24-hour care during weeklong outdoor camping trips with 12 children per week (ages 8 to 18), teaching survival skills, rock climbing, whitewater rafting, horseback riding, and cave camping.

TECHNICAL SKILLS

- **Software:** AFT Fathom, ANSYS Fluent, ANSYS Mechanical, SolidWorks, Onshape, OpenRocket, RocketPy
- **Machines:** Scanning Electron Microscope (SEM), Metal X 3D Printer, CNC, WaterJet, Lathe
- **Programming Languages:** MATLAB, Python

PROJECTS

Unmanned Aerial Vehicle (UAV)

- Engineered a custom UAV with 3D-printed parts to support customizability and quick repairs.
- Conducted ANSYS CFD simulation to validate conical surface performance.

Rocketry

- Built a two-stage rocket that reached over 2,500 feet and utilized my custom recovery system.